

Antonio Carmagnani

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PROFILE

Experience across sustainability consulting and the built environment, from ESG and human rights due diligence to climate risk work on internationally financed infrastructure. Recently completed an M.Arch. in Architecture Computation at The Royal Danish Academy, with a focus on data-driven approaches to sustainability. Hands-on skills span the UNGP and OECD due-diligence frameworks, climate hazard and risk assessment, and building structured analytical tools in Excel and Python. Motivated by work where sustainability, the built environment, and rigorous analysis intersect.

EXPERIENCE **Project Assistant in International Infrastructure Projects @ Sweco Danmark A/S**

Copenhagen, DK

02.2026 - 07.2026

- Authored climate risk assessments for major water supply projects following JASPERS and EBRD climate-proofing guidance, screening hazards against projections data and rating risk by exposure and vulnerability for each asset.
- Joined a field mission across Baku's water supply system, assessing asset condition and identifying technical, social, environmental, and human rights risks through site inspections and interviews.
- Engaged national water agencies and government counterparts during the field missions to align on rehabilitation priorities and collect technical and geographic project data.
- Built geospatial databases for water and infrastructure projects in QGIS, compiling georeferenced asset locations from agency data, satellite imagery, and open sources for socio-spatial analysis.
- Support consulting teams delivering internationally funded infrastructure projects in water, energy, and transport across emerging and developing economies, financed by development banks including the World Bank, EBRD, IFDK, KfW and EIB.

Intern in Responsible Business Conduct @ Ramboll Management Consulting

Copenhagen, DK

05.2025 - 08.2025

- Developed a client-facing overview of the firm's real estate ESG due diligence services, centered on building energy and decarbonization assessments aligned with EU building performance regulations.
- Built a fully automated risk-assessment system in Excel, consolidating risk data across supply chains into a single dashboard that updated salient risks dynamically.
- Created workshop and training materials on business and human rights for a corporate client, drawing on real-world cases to illustrate human rights risks, their business repercussions, and mitigation measures.
- Built a Human Rights Impact Assessment tool, mapping 40+ salient human rights risks across supply chains, with severity scoring to prioritize the most material risks.
- Conducted human rights due diligence for corporate clients, applying the UN Guiding Principles on Business and Human Rights and OECD Guidelines to assess salient risks across global supply chains.

Architect @ Isay Weinfeld Architects

São Paulo, BR

02.2023 - 08.2024

- Optimized building orientation and massing for energy performance and daylight using environmental simulation.
- Coordinated multidisciplinary inputs (structural, HVAC, electrical, lighting) into a single Revit model.
- Built a Grasshopper plugin in Python that generates bespoke staircases from a floor plan, cutting staircase modeling from days to around 20 minutes and enabling far more design iterations and geometric complexity.
- Delivered high-end residential, hotel, and mixed-use projects across South America, North America, Europe, the Middle East, and Asia at one of Brazil's most prominent architecture studios.

Trainee in Real Estate Product Development @ OSPA

São Paulo, BR

02.2022 - 10.2022

- Ran preliminary environmental and solar studies to inform massing and layout decisions.
- Ran early-stage financial feasibility on candidate sites, sizing investment and return against buildable area, unit mix, price per square metre, and parking to test viability.
- Produced 15+ feasibility studies for mixed-use and residential developments across Brazil and the US (São Paulo, Rio de Janeiro, Porto Alegre, Miami) for developer clients.

Co-Founder @ VIZU

São Paulo, BR

02.2022 - 10.2022

- Co-founded and ran an architectural visualization studio, managing 10+ clients, project scoping and delivery, and hiring staff.
- Produced architectural renders, humanized plans, and visualizations to communicate residential and commercial designs to investors and end buyers.

Architect @ Felipe Hess Architects

São Paulo, BR

07.2021 - 02.2022

- Ran environmental and daylight studies to inform early design decisions.
- Developed high-end residential projects in São Paulo and Rio de Janeiro from preliminary studies through technical documentation, working in Revit, AutoCAD, and SketchUp.

08.2020 - 07.2021

- Assessed the environmental and energy performance of buildings the bank was acquiring or leasing, reviewing consumption data, efficiency ratings, and certifications as part of internal due diligence.
- Supported technical review of new office towers under construction at the bank's São Paulo headquarters campus, evaluating design documentation for compliance and environmental performance.
- Built a VBA-powered Excel dashboard tracking the bank's corporate furniture inventory across locations, replacing a manual process and speeding up procurement for new-building fit-outs.
- Reviewed technical documentation from architecture and engineering firms prior to construction authorization, checking drawings against building, accessibility, and fire-safety norms (NBR standards).

EDUCATION**M.Arch. in Architecture Computation @ The Royal Danish Academy****Copenhagen, DK***07.2024 - 07.2026***Semester Programme in Building Technology @ Architecture Association****London, UK***07.2022 - 01.2023***Summer School in Parametric Design @ Architecture Association****London, UK***06.2019 - 07.2019***B.Arch. in Architecture and Urbanism @ Mackenzie University****São Paulo, BR***07.2018 - 12.2023***PROJECTS****Embodied Carbon Toolkit****Copenhagen, DK**

- Collaborated with a Copenhagen-based LCA startup to design and deliver a Revit-based toolkit for embodied carbon calculations, integrating the platform's GWP values for A1-A3 and A4 stages.
- Built a pre-configured Revit schedule that computes embodied carbon with built-in documentation for adoption by architects.

Real Estate Development Recommendation Tool @ The Royal Danish Academy**Copenhagen, DK**

- Built a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by integrating financial, social, sustainability, and geospatial data.
- Designed a user interface allowing different stakeholders to adjust impact drivers and weighting, with predictive outputs aligned to ESG priorities and local development conditions.

Rammed Blue Biomass & Computational Optimization @ The Royal Danish Academy**Copenhagen, DK**

- Developed a new construction technique and material system using blue biomass (seashells and algae), applying LCA metrics and parametric simulations to optimize structural performance and environmental impact.
- Project was selected as a semester highlight and served as the material foundation for the school's installation at the Venice Architecture Biennale 2025, presented in collaboration with CITA, GXN, and 3XN at the Arsenale.

Master's Thesis, Computational Urban Analysis for Intensification Strategy @ The Royal Danish Academy**Copenhagen, DK**

- Developed a computational framework to identify underused spaces in the historic city center of São Paulo, Brazil, combining urban morphology analysis, land use data, and social vulnerability indicators.
- Built an algorithm that uses urban data, including building typology, use patterns, regulatory conditions, and building use, to generate programmatic and morphological typologies for urban intensification, aligned with the city's existing urban policies.
- Integrated multiple geospatial datasets in QGIS to map spatial relationships between buildings, zoning conditions, and social context at the neighborhood scale.

SKILLS

- Data & Analysis: Python, QGIS, SQL, Power BI, Excel / VBA, HTML
- Digital Tools: Rhino + Grasshopper, Revit, AutoCAD, Adobe Suite (InDesign, Illustrator, Photoshop)
- ESG & Sustainability: CSRD, Double Materiality, EU Taxonomy, UNGPs, OECD Guidelines, HRDD, HRIA, LCA (Life Cycle Assessment)
- Real Estate & Urban: Real estate feasibility modeling, Zoning and land use regulatory analysis, Urban morphology and spatial analysis
- Languages: English (fluent), Portuguese (native), Spanish (fluent), Danish (limited working proficiency), Italian (limited working proficiency)